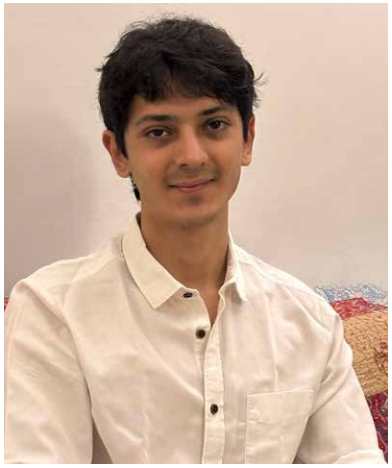
Zai Buragohain

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Mobile (INDIA): +91 9910721277



EDUCATION

MEng (Hons)
Integrated Design Engineering,
The University of Bath, UK
2026

ABOUT ME

I'm deeply interested in both the technical and the artistic aspects of design, with a focus on human centred design and product development. I aim to create functional designs that feel good to use.

I believe that paying close attention pays off. Questioning why a client kicked the back of their bike led to discovering an issue they had with bike maintenance. What initially seemed like an irrelevant question became the main inspiration for the Mudguard Brush design.

I hope to further refine my creativity and curiosity.

SKILLS

DESIGN
Human Centred Design thinking | Concept Generation | Traditional & Digital Sketching | Illustration

PROTOTYPING
Rapid Prototyping using various materials | 3D Printing | Laser Cutting | Machining | Soldering | Digital Prototyping Simulations

ENGINEERING
Foundation in Mechanical Engineering | Problem Solving | Systems thinking | Adaptability

INTERPERSONAL SKILLS
Teamwork & Cooperation | Approachable | Adaptable

SOFTWARE

- Rhino 3D + Grasshopper
- AutoCAD
- Fusion 360
- ProtoPie
- Figma
- Procreate
- Python
- Miro
- Microsoft Office Suite

Modular Monitor Mount

SUMMER INTERNSHIP – **FOSTER + PARTNERS, LONDON** | 2/6/2025 – 29/8/2025

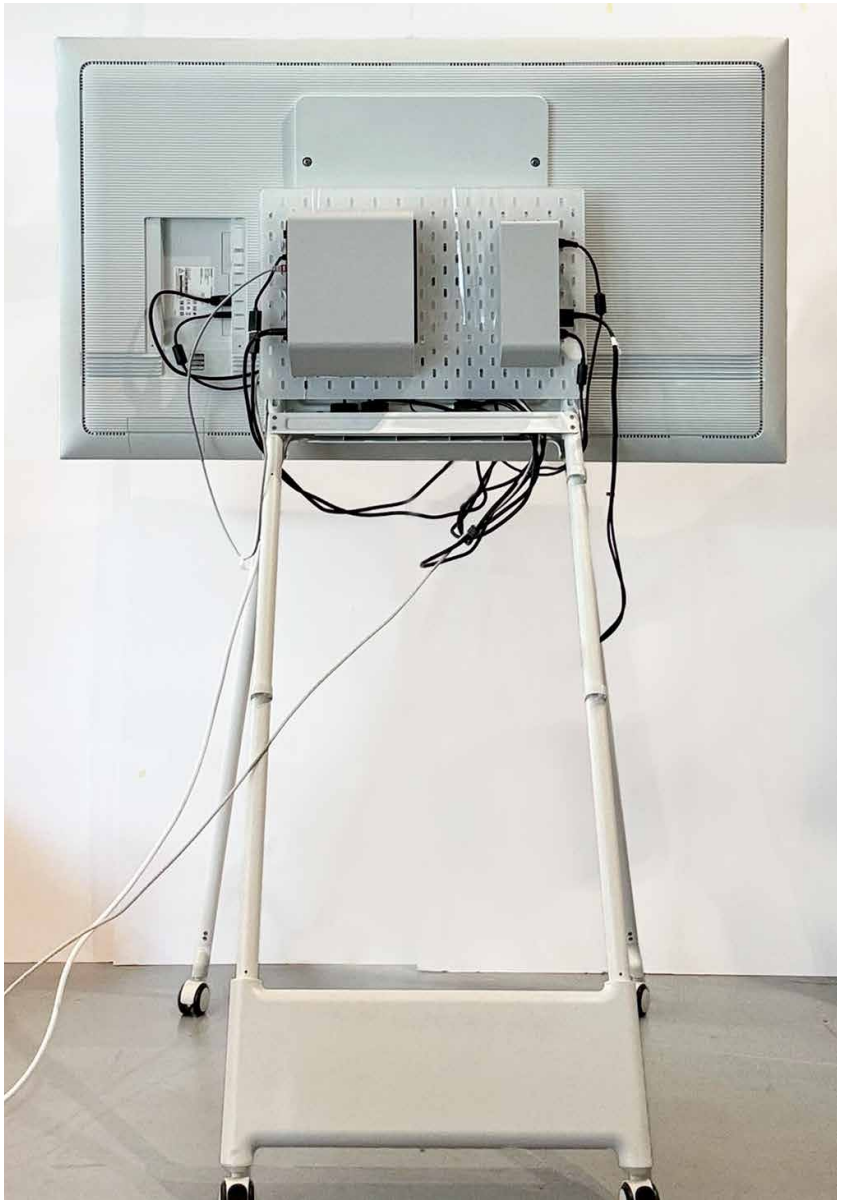
In the last two weeks of my internship at Foster + Partners, I worked for the Industrial Design team to design a modification for a presentation monitor in use at the office.

A PC, power brick and cables for a presentation monitor were stored in a wooden cart. Adjusting the position of the monitor for a presentation required reaching down to move the cart as well. Thus, the solution required a method of attaching the electronic components directly onto the monitor.

The presented solution is incomplete. The model for the second power brick's mount was ready but could not be printed within the timeframe. Additionally, cable management solutions were necessary to complete the design.



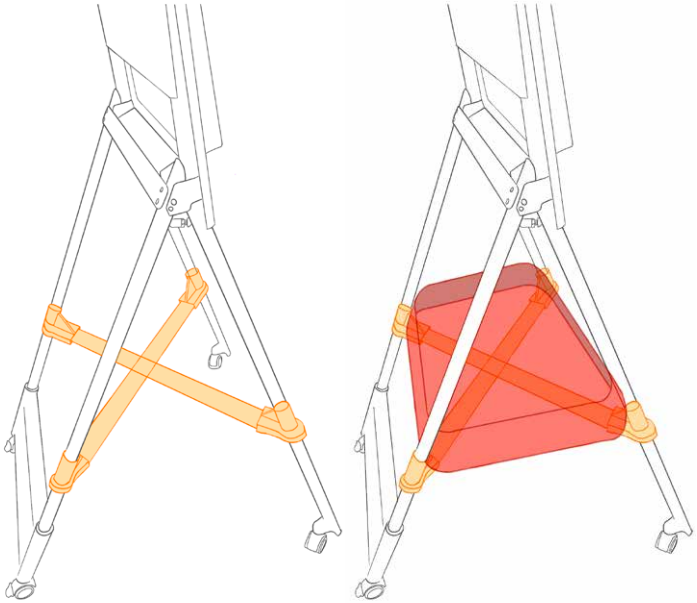




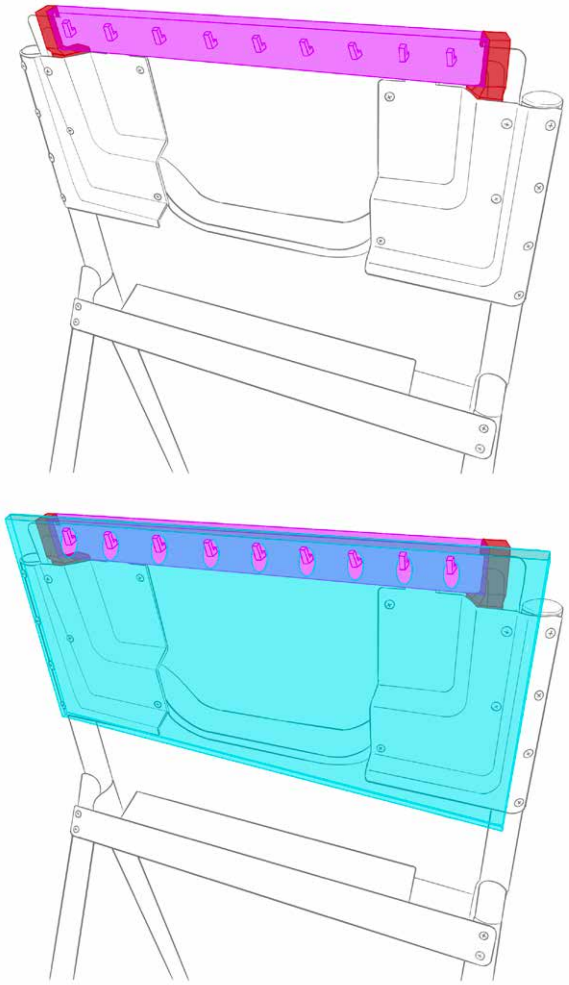
CONCEPT SKETCHES

1. Base Tray

Concept 1 had good storage capacity but required clamps with secure fastening methods. Attaching/detaching the base tray would be cumbersome.

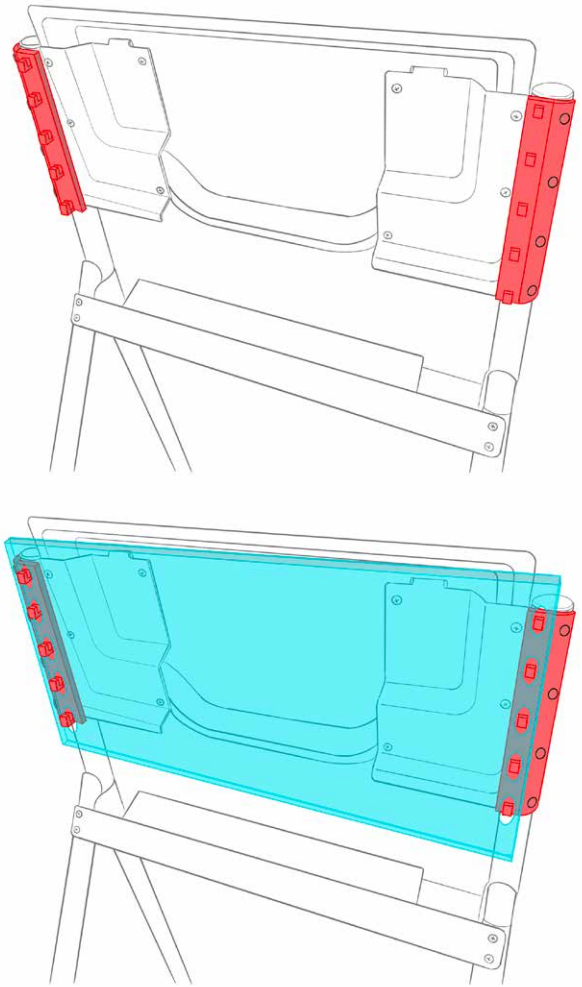


2. Top Pegboard Mount



The pegboard in Concept 2 is only constrained to the top lip of the back frame. When the monitor is moved, this would introduce strain on the 3D printed parts and break them apart.

3. Side Pegboard Mount

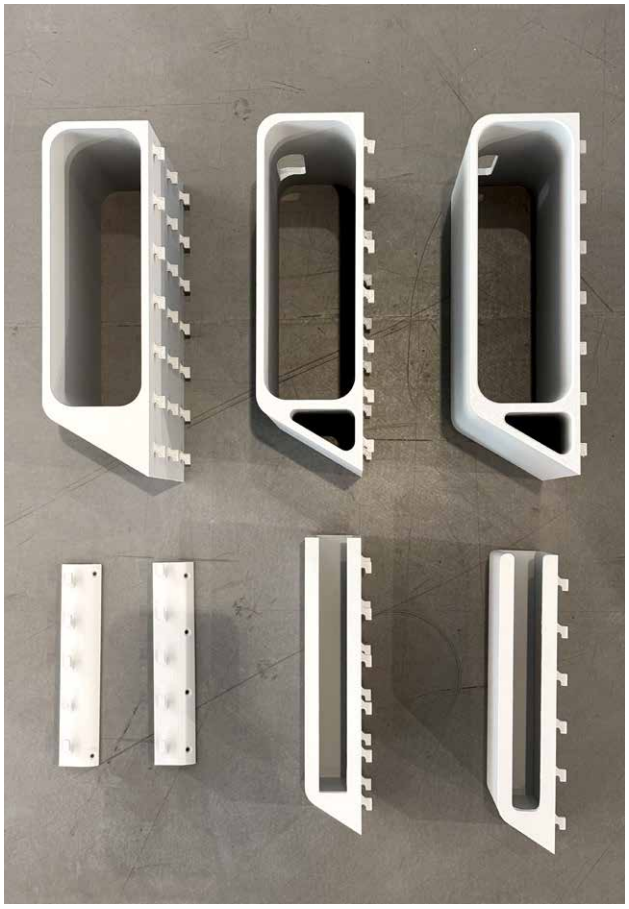


Concept 3 uses the pre-existing bolt placements to secure the pegboard mounts to ensure that it remains constrained. Hence, Concept 3 was selected for the prototype.

DESIGN REQUIREMENTS

1. A stable attachment method
2. Storage for multiple heavy components
3. Accommodation for new hardware
4. Secure objects on mount while in motion
5. Easily removable / attachable solution
6. Cable management
7. All without damaging the base structure

COMPONENTS AND ASSEMBLY





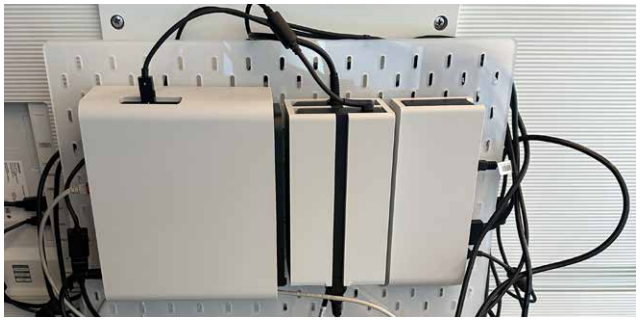


After 3D-printing and laser-cutting, the parts were tested for their fit and iterated upon where necessary.

The components were then installed on the monitor.







1. Base Monitor



2. Back cover removed



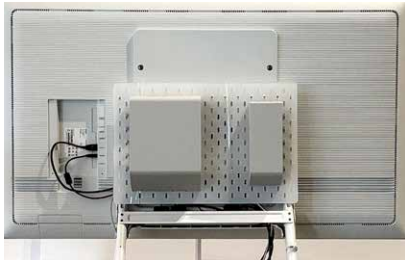
3. Screw Mounts Added



4. Pegboard Mounted



5. Component Mounts Added



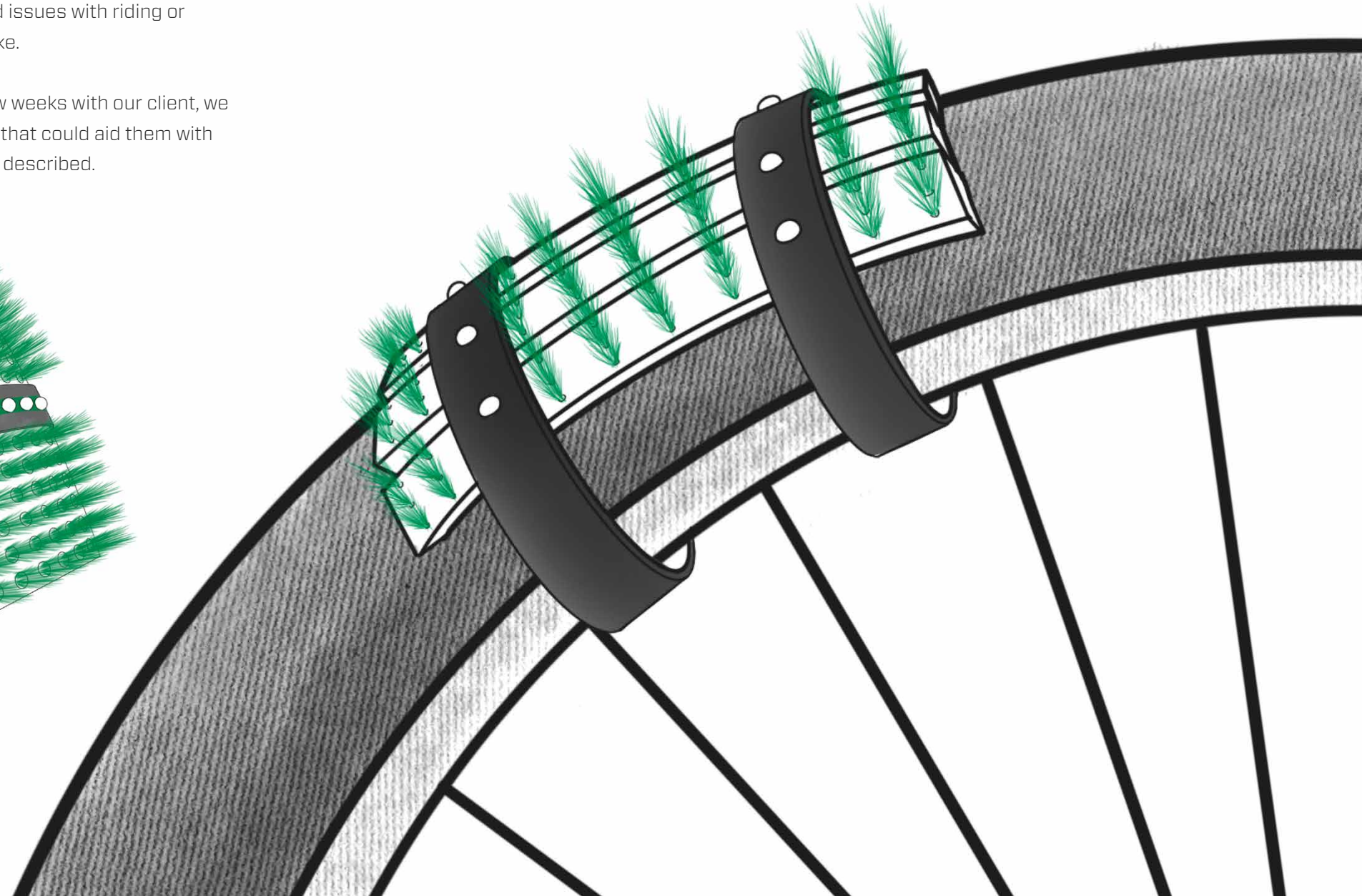
Mudguard Brush

UNIVERSITY OF BATH | **GROUP PROJECT** | 30/9/2024 – 30/10/2024

As a part of a university project, we met with clients who had issues with riding or maintaining their bike.

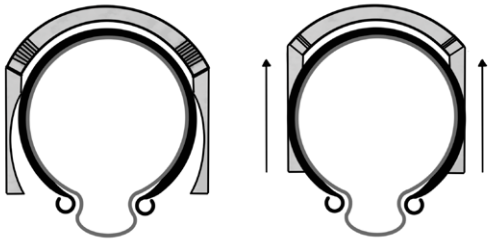
At the end of our few weeks with our client, we designed a product that could aid them with the challenges they described.



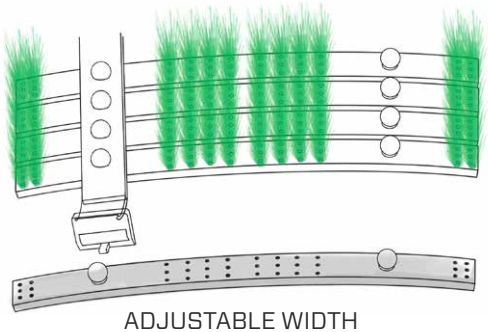




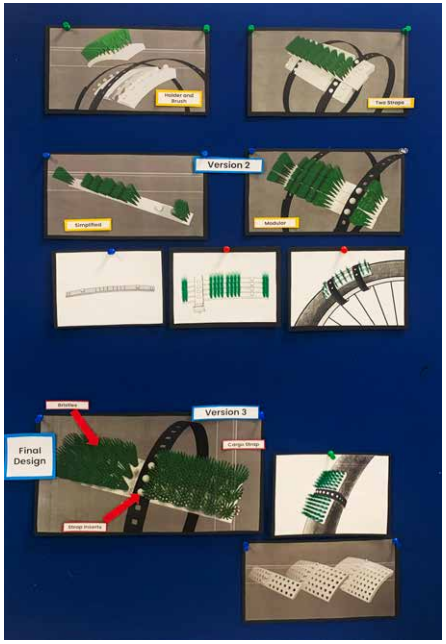
The client we were designing for claimed they had no issues with riding their bike. They owned a cargo bike with a cover on the back wheel. While watching them begin to ride, I noticed that they kicked the back cover before pedalling. When asked about this, they mentioned that the under side of the mudguard would collect dirt and is difficult to clean because of the cover on the back wheel.
This inspired the Mudguard Brush idea.



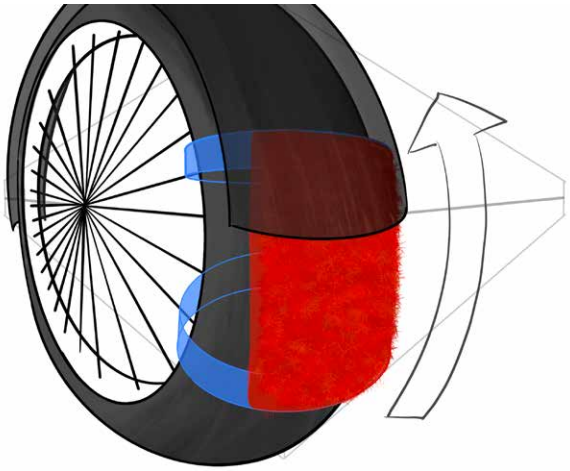
CLAMP MECHANISM



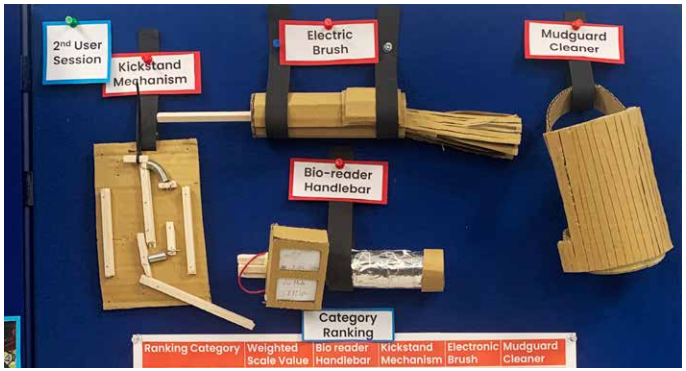
ADJUSTABLE WIDTH

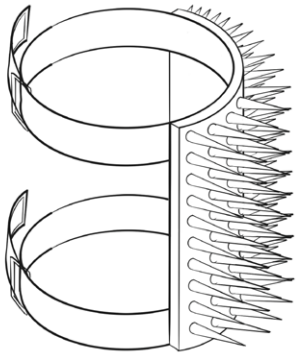


The idea was to strap a brush onto the wheel of the bike and rock it back and forth to clean the inside of the inaccessible mudguard.
In the following user session, 4 concept ideas were presented to the clients. Based on the feedback, the mudguard brush was chosen to be taken into the next phase of design.

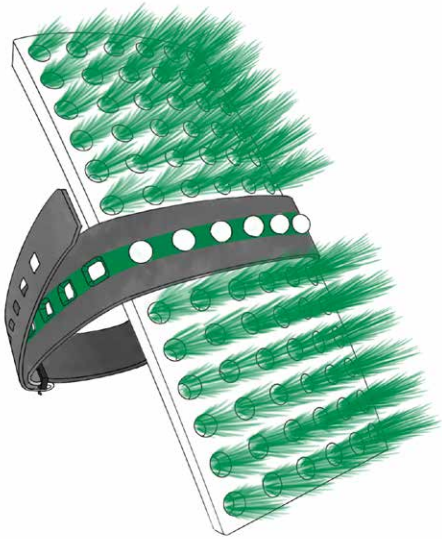


After exploring attachment methods, and adaptable, modular designs, we were able to converge on a functional and feasible product that was received well by the client.







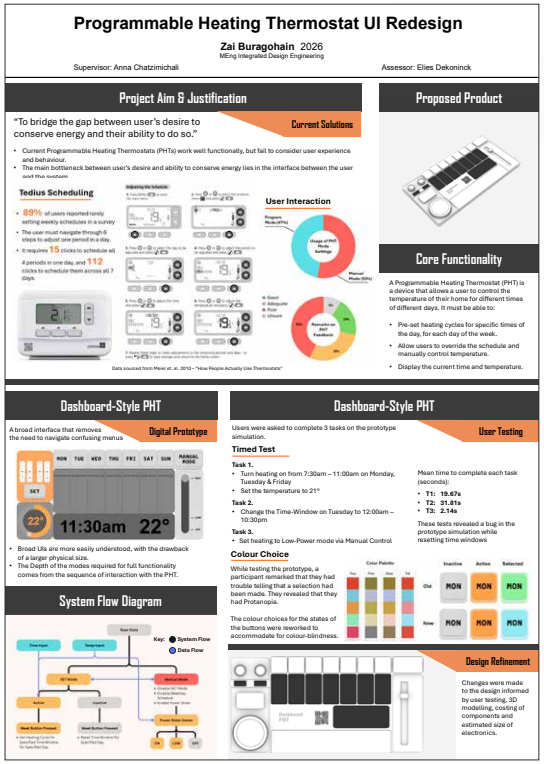


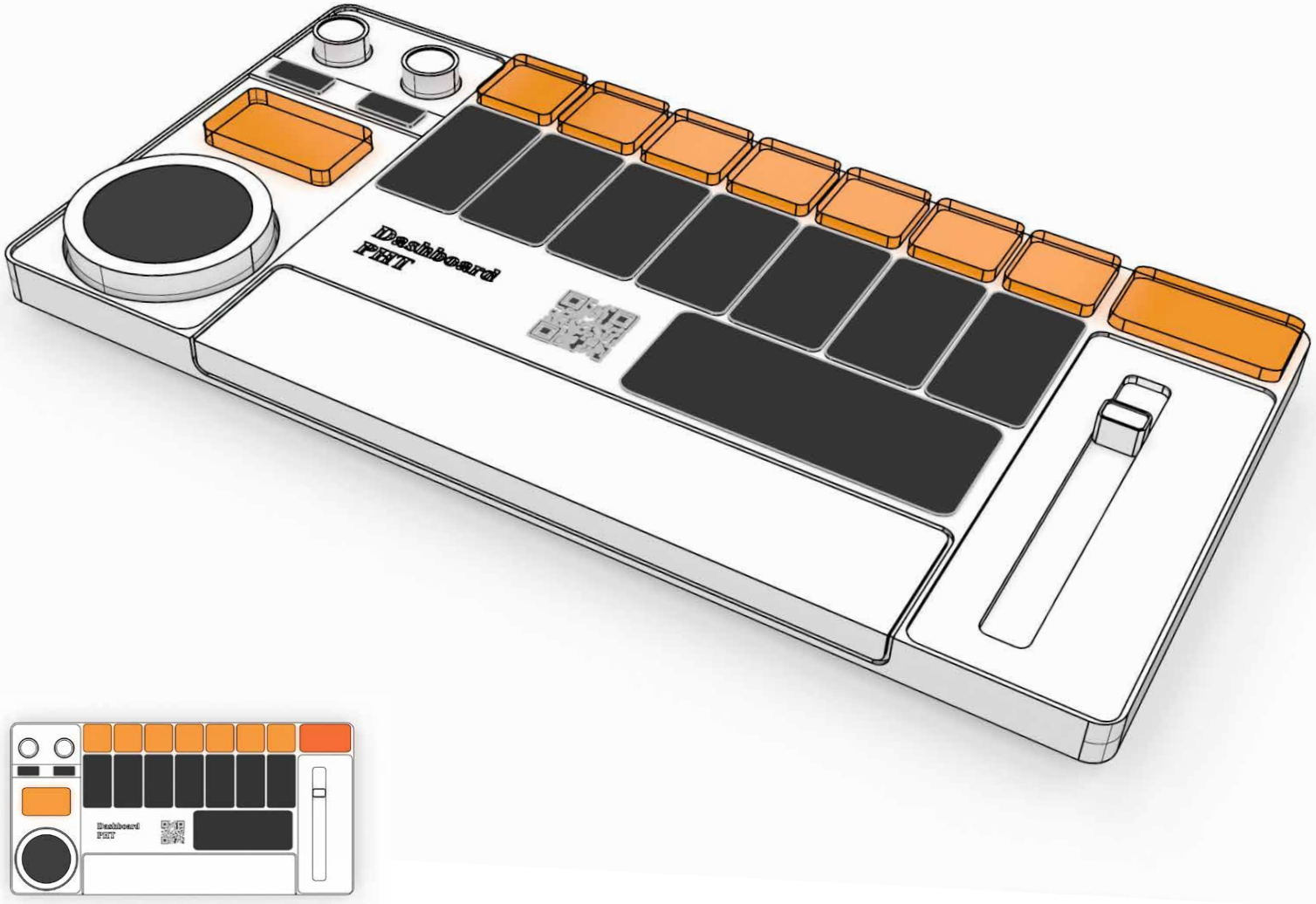
Dashboard-PHT

UNIVERSITY OF BATH | **INDIVIDUAL PROJECT** | 3/10/2025 – 21/5/2026

A Programmable Heating Thermostat UI Redesign, focusing on usability and Human Centred Design.

This design moves away from deep menus that require a sequence of button presses to get to a specific setting and instead presents all modes and menus at the base-level.





Programmable Heating Thermostat UI Redesign

Project Aim & Justification

"To bridge the gap between user's desire to conserve energy and their ability to do so."

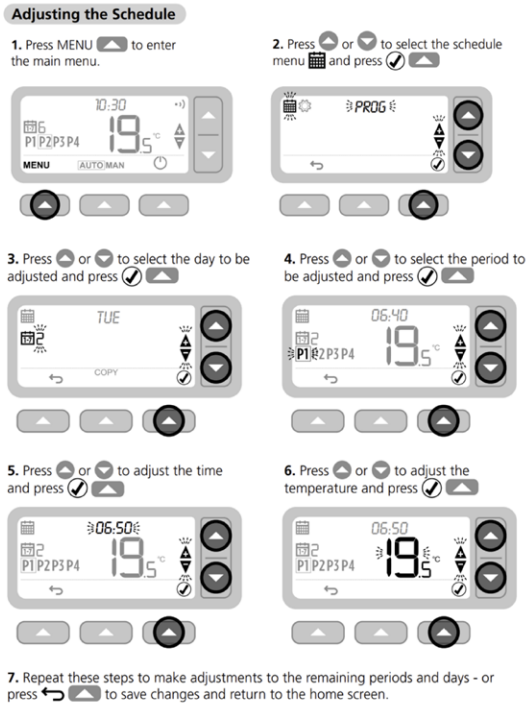
Current Solutions

- Current Programmable Heating Thermostats (PHTs) work well functionally, but fail to consider user experience and behaviour.
- The main bottleneck between user's desire and ability to conserve energy lies in the interface between the user and the system.

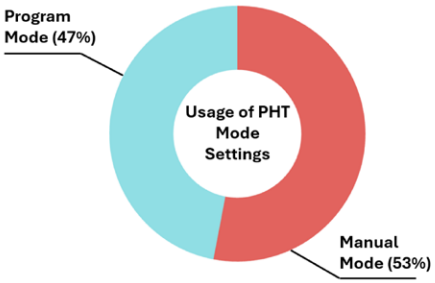
Tedious Scheduling

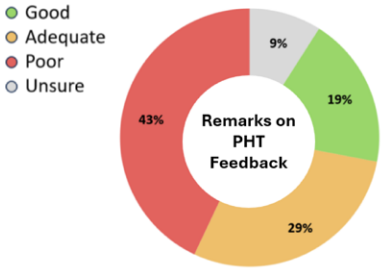
- **89%** of users reported rarely setting weekly schedules in a survey
- The user must navigate through 6 steps to adjust one period in a day.
- It requires **15** clicks to schedule all 4 periods in one day, and **112** clicks to schedule them across all 7 days.





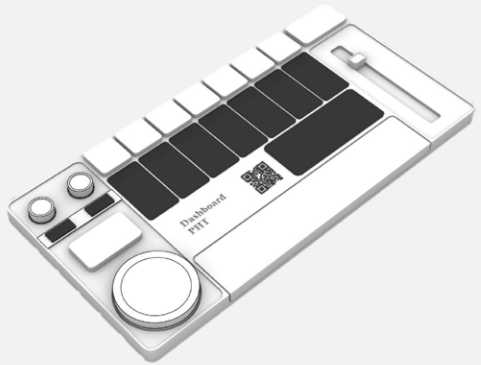
User Interaction





Data sourced from Meier et al. 2010 – "How People Actually Use Thermostats"

Proposed Product



Core Functionality

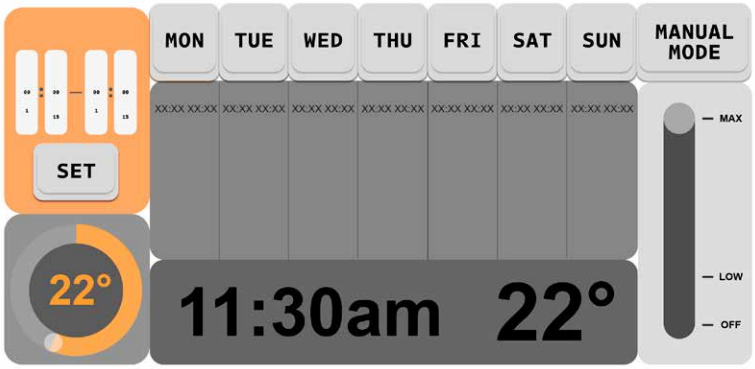
A Programmable Heating Thermostat (PHT) is a device that allows a user to control the temperature of their home for different times of different days. It must be able to:

- Pre-set heating cycles for specific times of the day, for each day of the week.
- Allow users to override the schedule and manually control temperature.
- Display the current time and temperature.

Dashboard-Style PHT

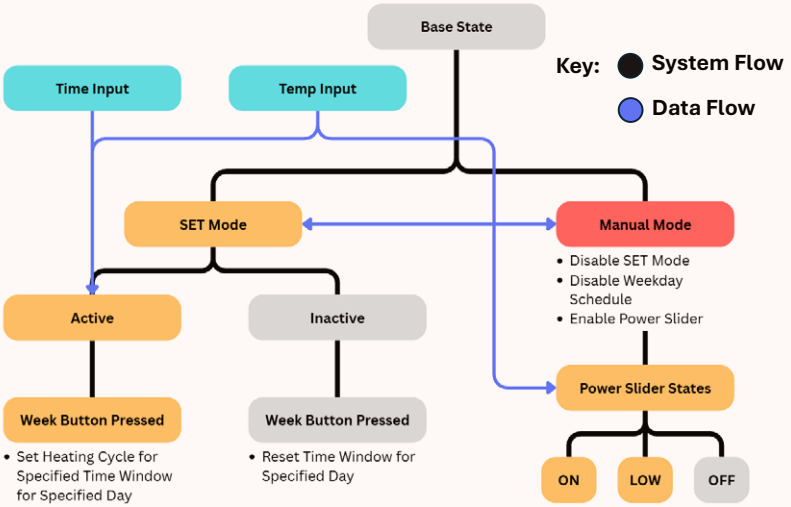
Digital Prototype

A broad interface that removes the need to navigate confusing menus



- Broad UIs are more easily understood, with the drawback of a larger physical size.
- The Depth of the modes required for full functionality comes from the sequence of interaction with the PHT.

System Flow Diagram



Dashboard-Style PHT

User Testing

Users were asked to complete 3 tasks on the prototype simulation.

Timed Test

Task 1.
- Turn heating on from 7:30am – 11:00am on Monday, Tuesday & Friday
- Set the temperature to 21°

Task 2.
- Change the Time-Window on Tuesday to 12:00am – 10:30pm

Task 3.
- Set heating to Low-Power mode via Manual Control

Mean time to complete each task (seconds):

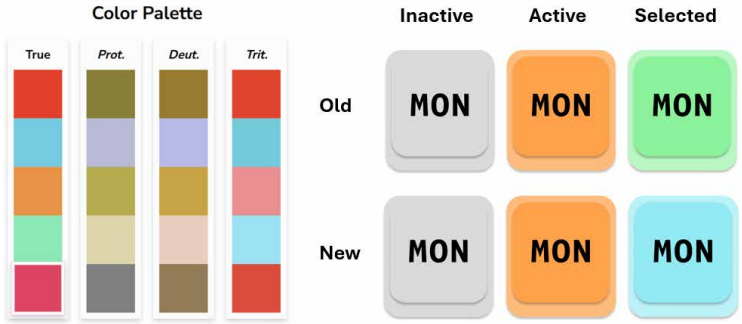
- **T1: 19.67s**
- **T2: 31.81s**
- **T3: 2.14s**

These tests revealed a bug in the prototype simulation while resetting time windows

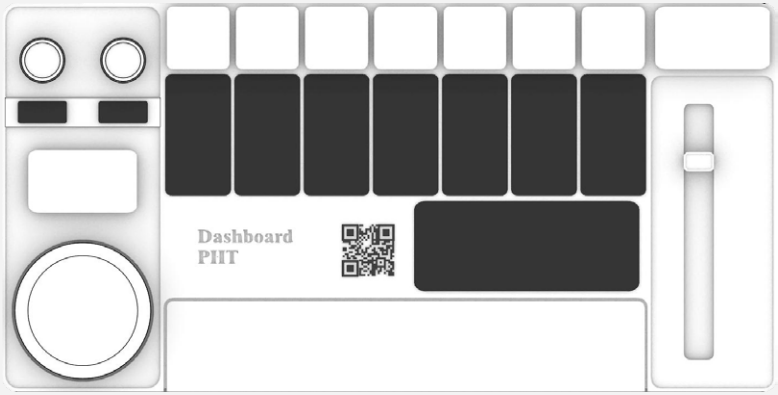
Colour Choice

While testing the prototype, a participant remarked that they had trouble telling that a selection had been made. They revealed that they had Protanopia.

The colour choices for the states of the buttons were reworked to accommodate for colour-blindness.



Design Refinement

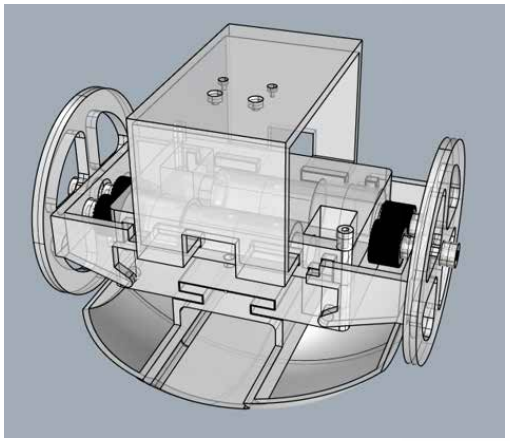


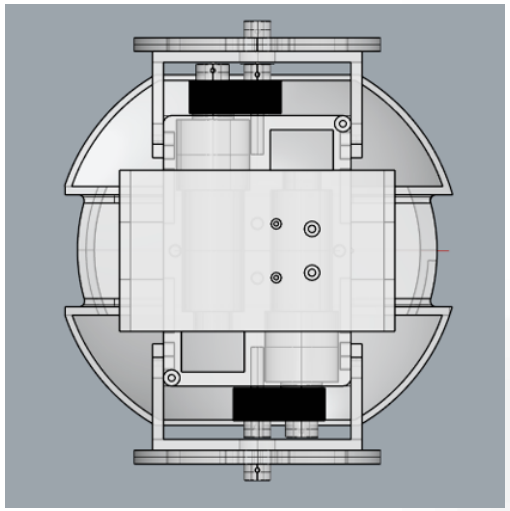
Changes were made to the design informed by user testing, 3D modelling, costing of components and estimated size of electronics.

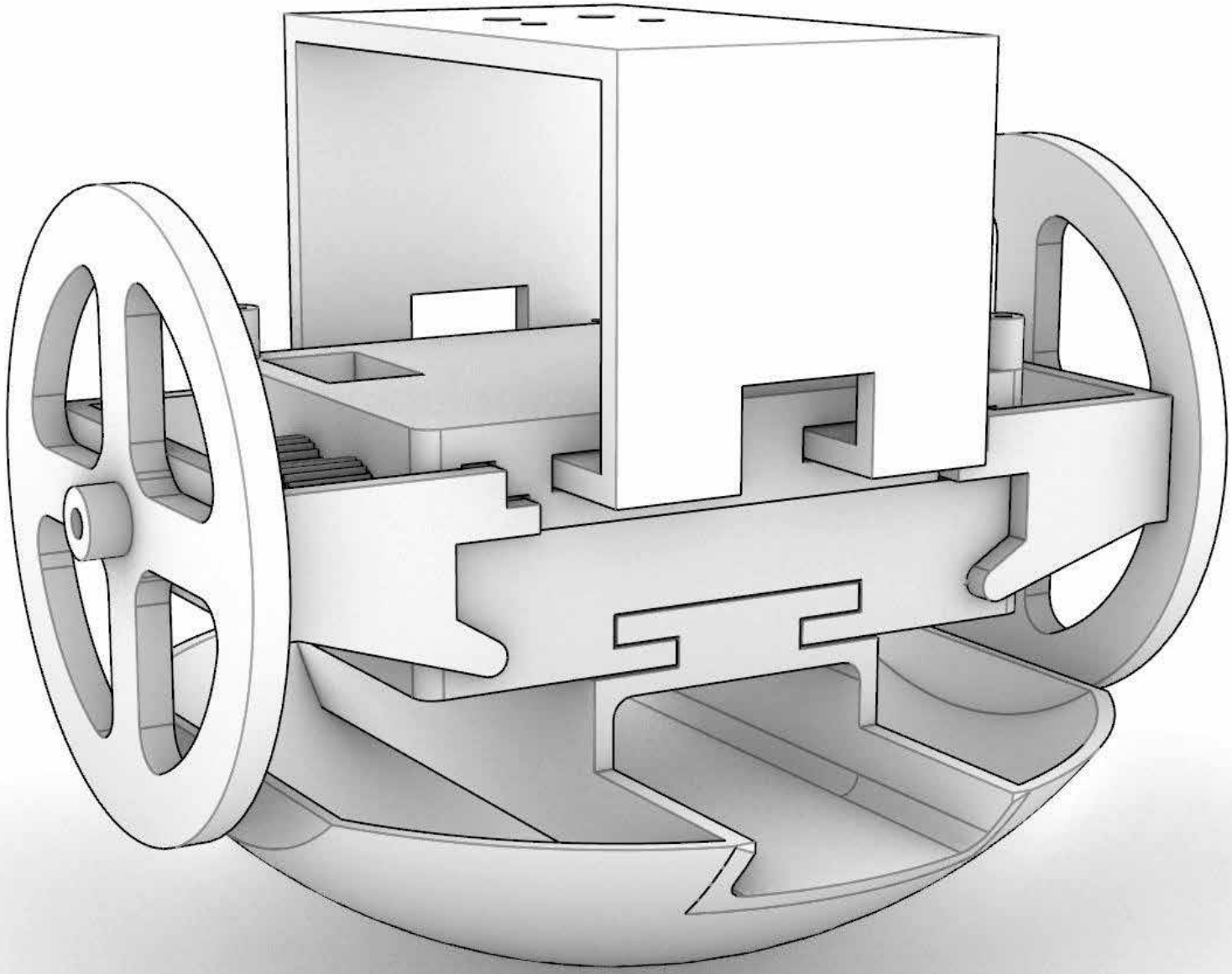
Encased Semi-Autonomous Robot

UNIVERSITY OF BATH | **MECHANICAL DESIGN LEAD** – GROUP PROJECT | 1/10/2025 – 5/11/2025

A Semi-Autonomous robot that could fit within a 195mm ball was built to go from point A to B in an arena while avoiding obstacles.





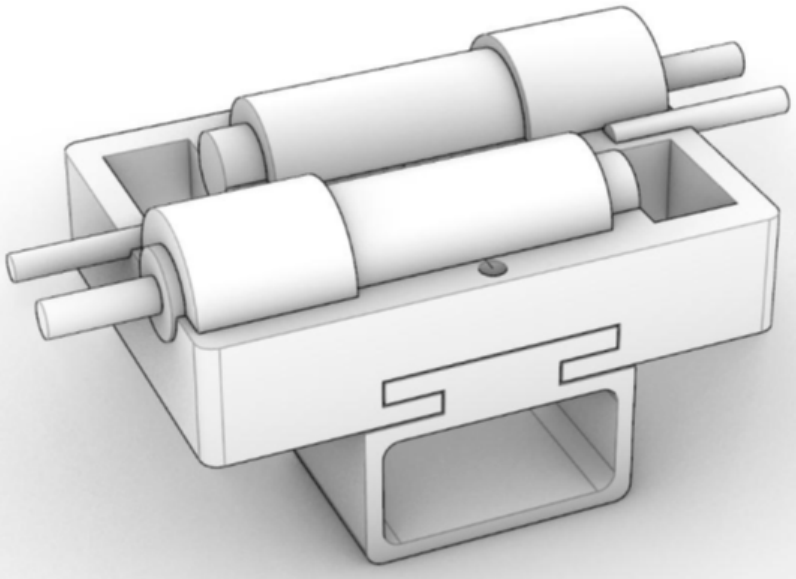


PROTOTYPES AND ITERATIONS

PROBLEM: How does one steer from inside a spherical shell?

SOLUTION: The idea was to turn the robot inside its shell. It was tested with a Lego mockup. Precise movement with only 4 modes makes the robot easier to drive from a control perspective.



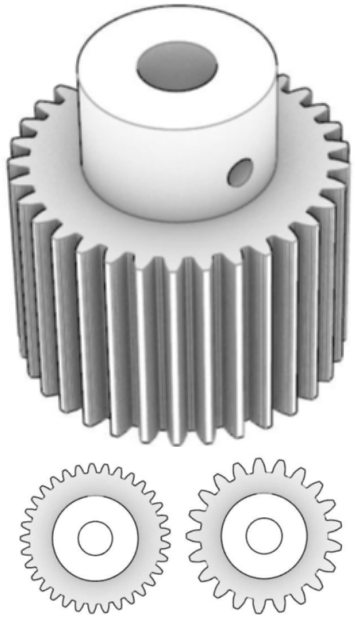


PROBLEM: Motors are too large to fit in line with one another as in the Lego mockup.

SOLUTION: Set them up side by side and use a gearbox with a central shaft to constrain the axis of rotation.

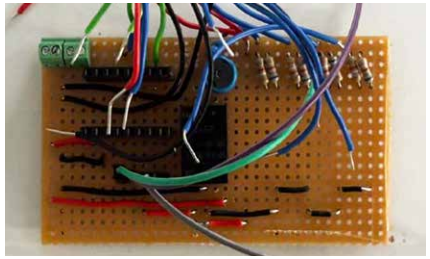
PROBLEM: The gears slipped and were unreliable for the translation of torque.

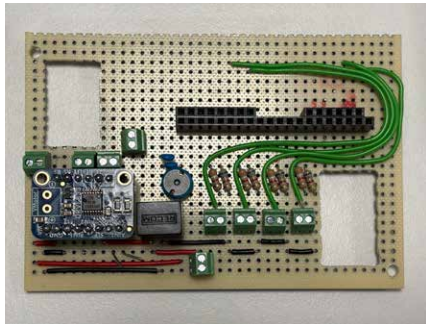
SOLUTION: Reducing the number of teeth on each gear led to better meshing and therefore better translation of torque.



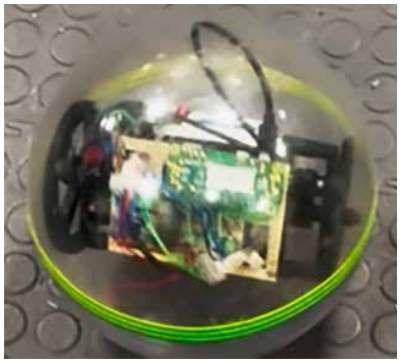
PROBLEM: The circuit was too small to be secured onto the standoffs on the main chassis and had confusing wiring.

SOLUTION: Neatly soldered the circuitry on a larger stripboard. Cut channels out of the stripboard for routing cables from the battery.

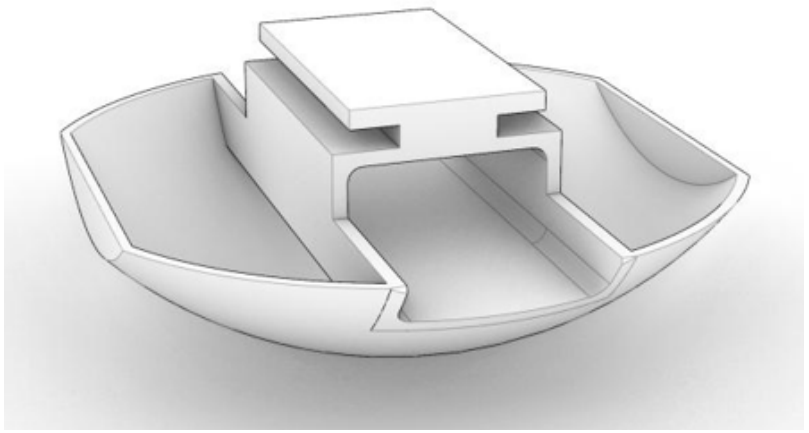




PROTOTYPES AND ITERATIONS





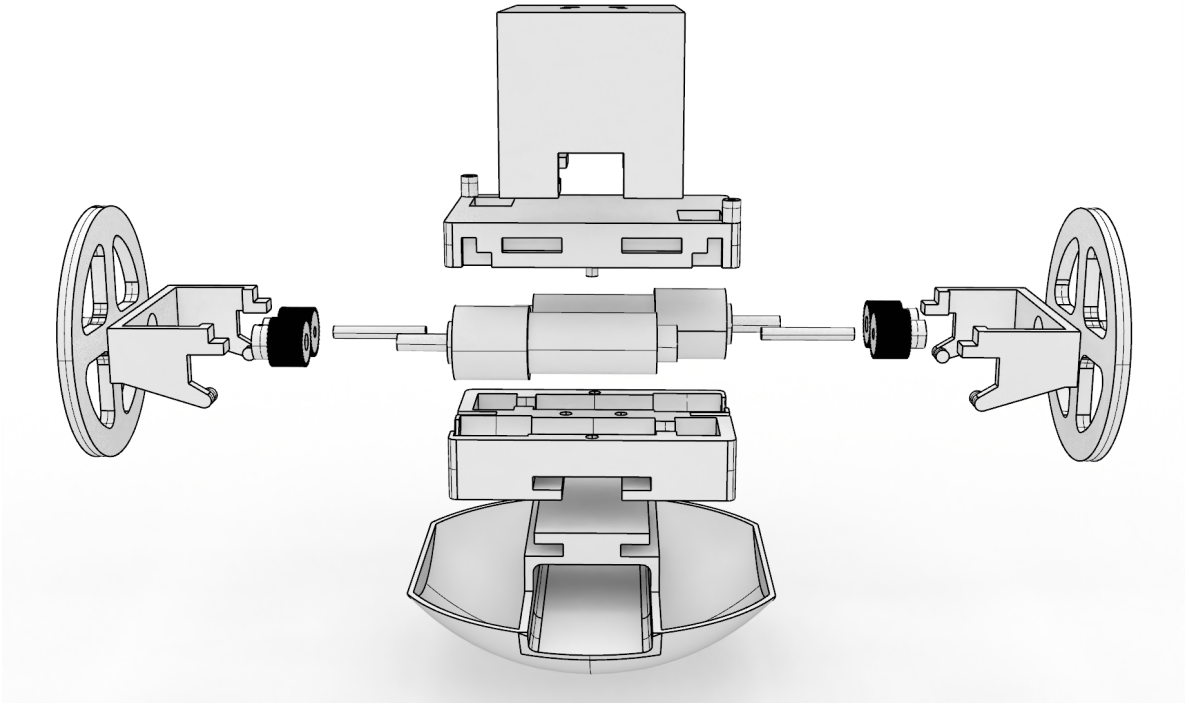


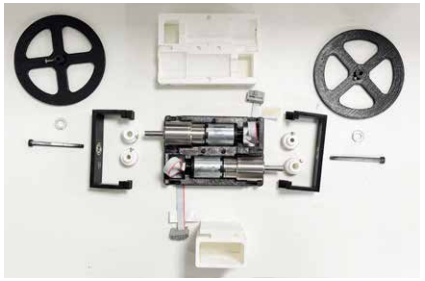
PROBLEM: The motors were the heaviest components and sat at the centre of the ball. Instead of rolling the ball forward, the robot backflipped inside it.

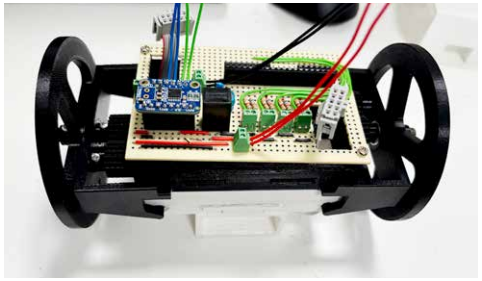
SOLUTION: A redesigned battery cubby carrying steel ball counterweights dropped the centre of mass low enough for stable movement and increased traction on the wheels.



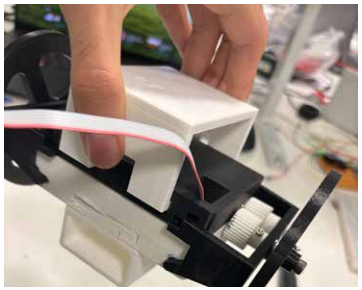
ASSEMBLY

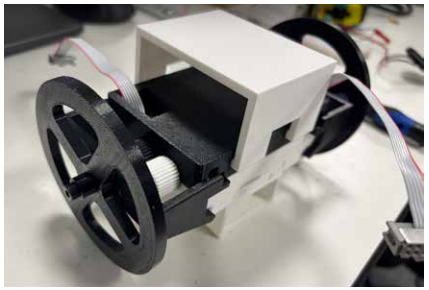


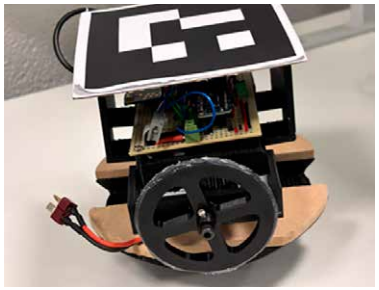












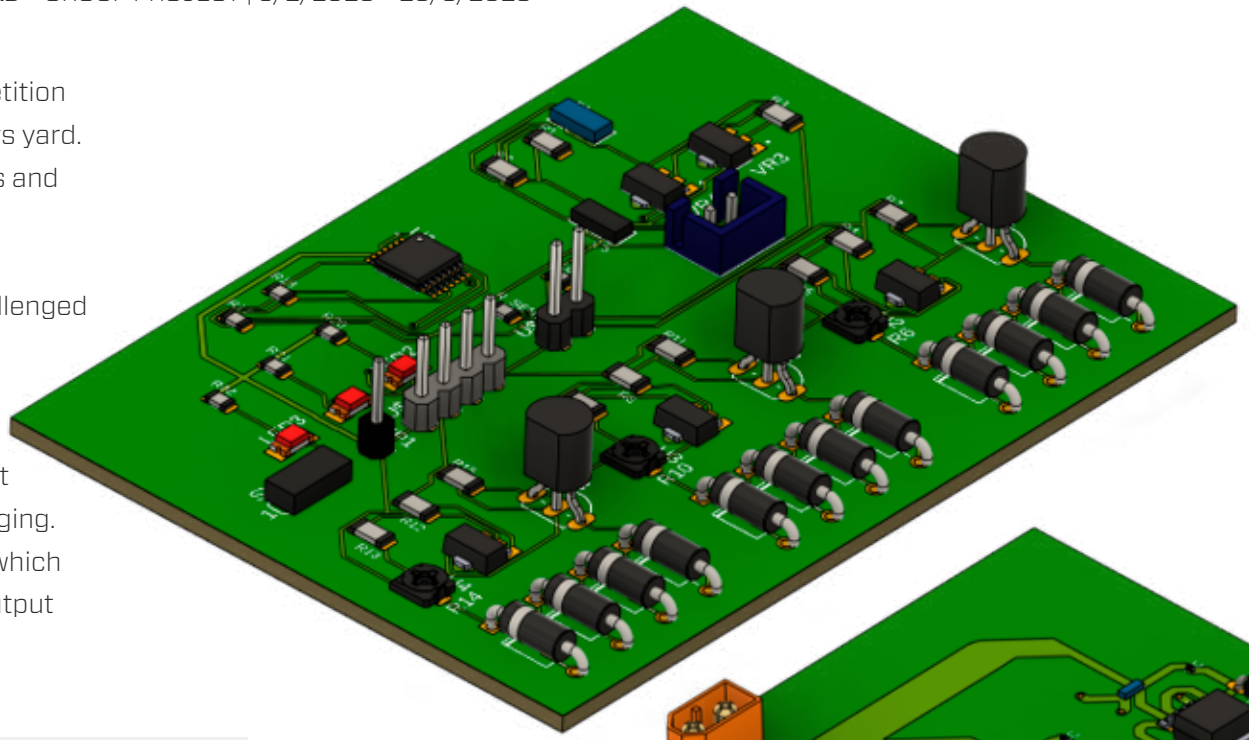
Power Management Subsystem

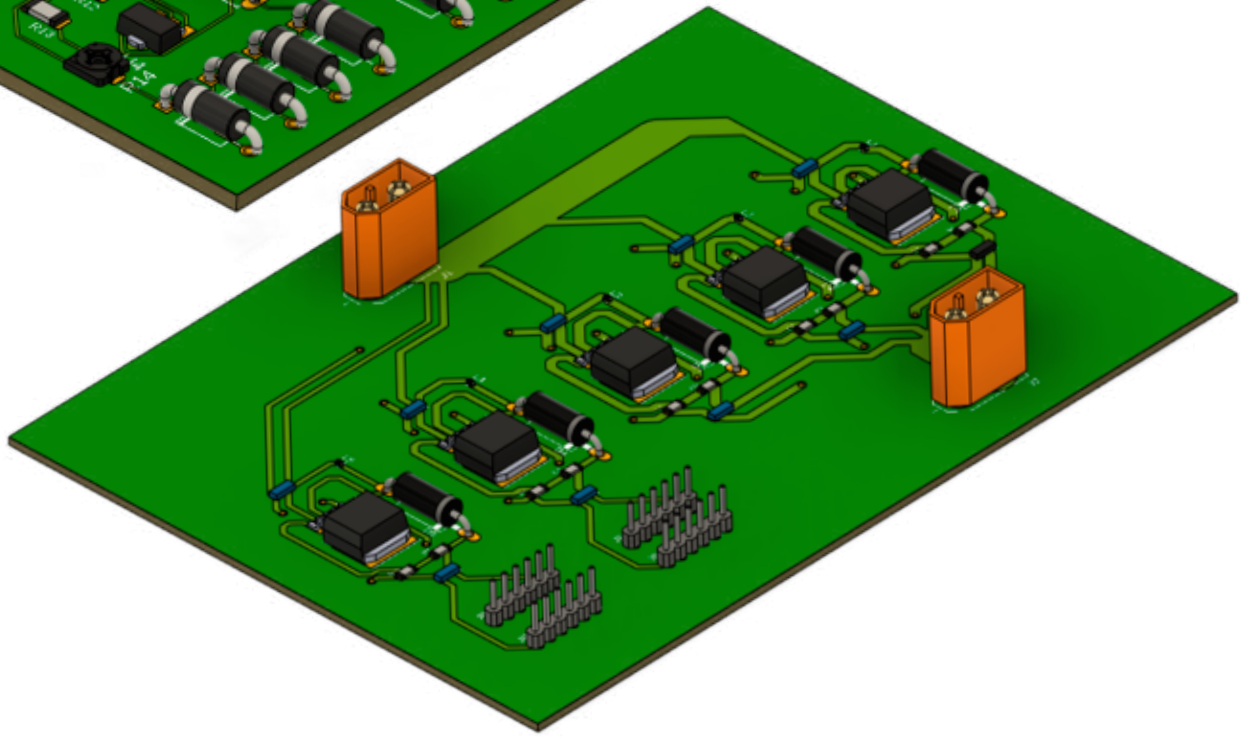
UNIVERSITY OF BATH | **POWER MANAGEMENT LEAD** – GROUP PROJECT | 3/2/2025 – 28/5/2025

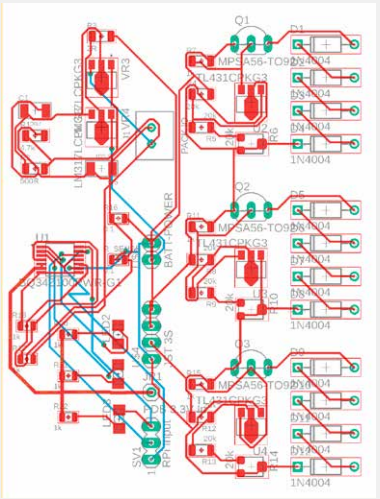
The UKSEDS Olympus Rover Trials is a competition where students build rovers to navigate a Mars yard. The rover had to manoeuvre around obstacles and scan several QR codes around the arena.

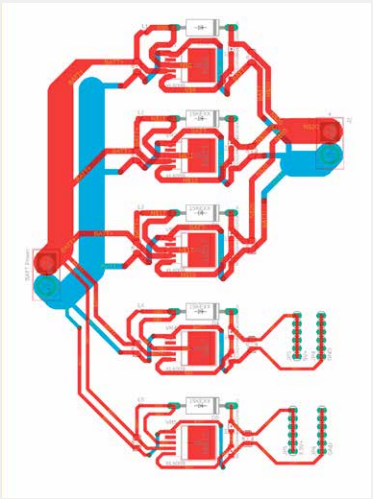
Although electronics is not my specialty, I challenged myself by taking on the role of the Power Management Lead.

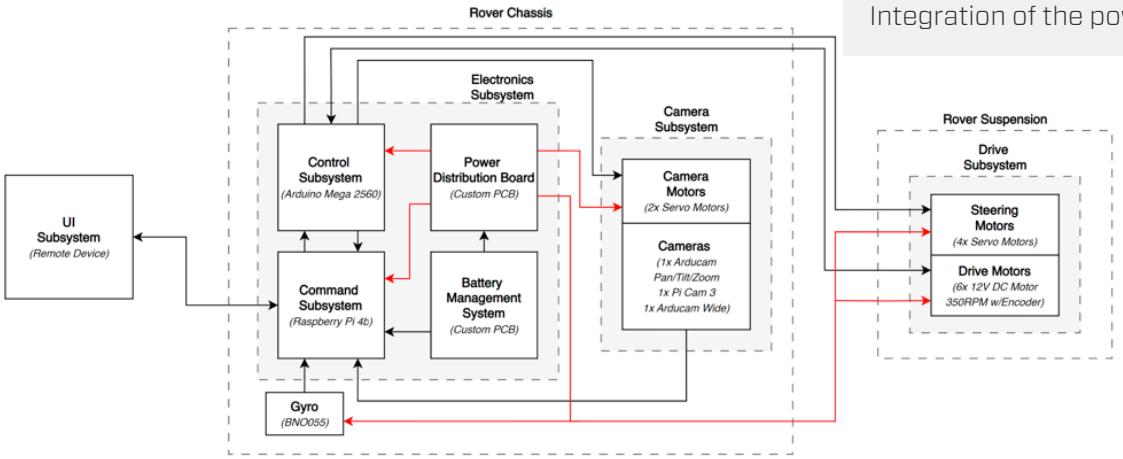
I designed a battery management system that regulates the battery and allows for safe charging. I also designed the power distribution board, which centralised the electronics and was able to output power to every other subsystem in the rover.



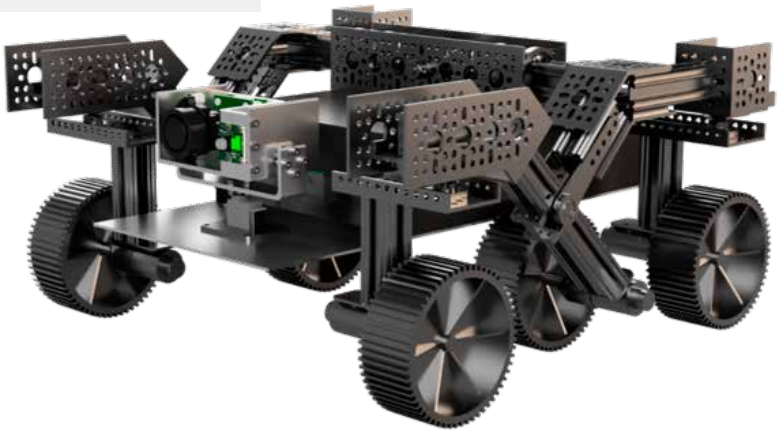


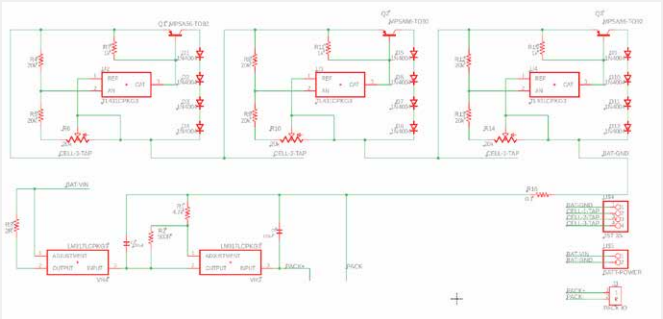


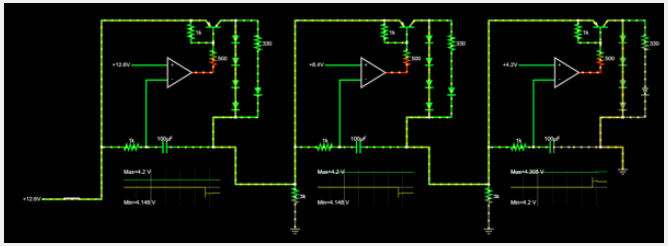




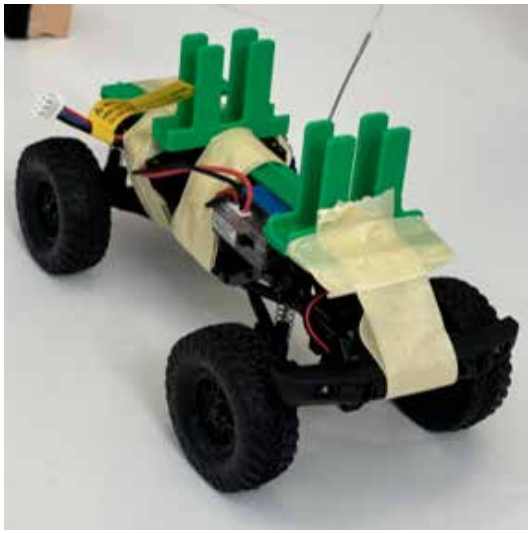
Integration of the power management subsystem.







This is a schematic for a 3-Cell Lithium Polymer Battery Management System (BMS). The design is validated by simulating the BMS in charging mode and noting that the voltage across the batteries (capacitors used in simulation) remains below 4.2V.





We needed to know what type of camera the rover required to manoeuvre around the obstacles in the mars yard. We used two phones taped to an RC car, a mocked-up yard with a QR code hidden from the driver, who could only navigate using the live feed. This allowed us to change the camera zoom settings to figure out the best balance between Field Of View and object image distortion to be able to scan the QR codes.



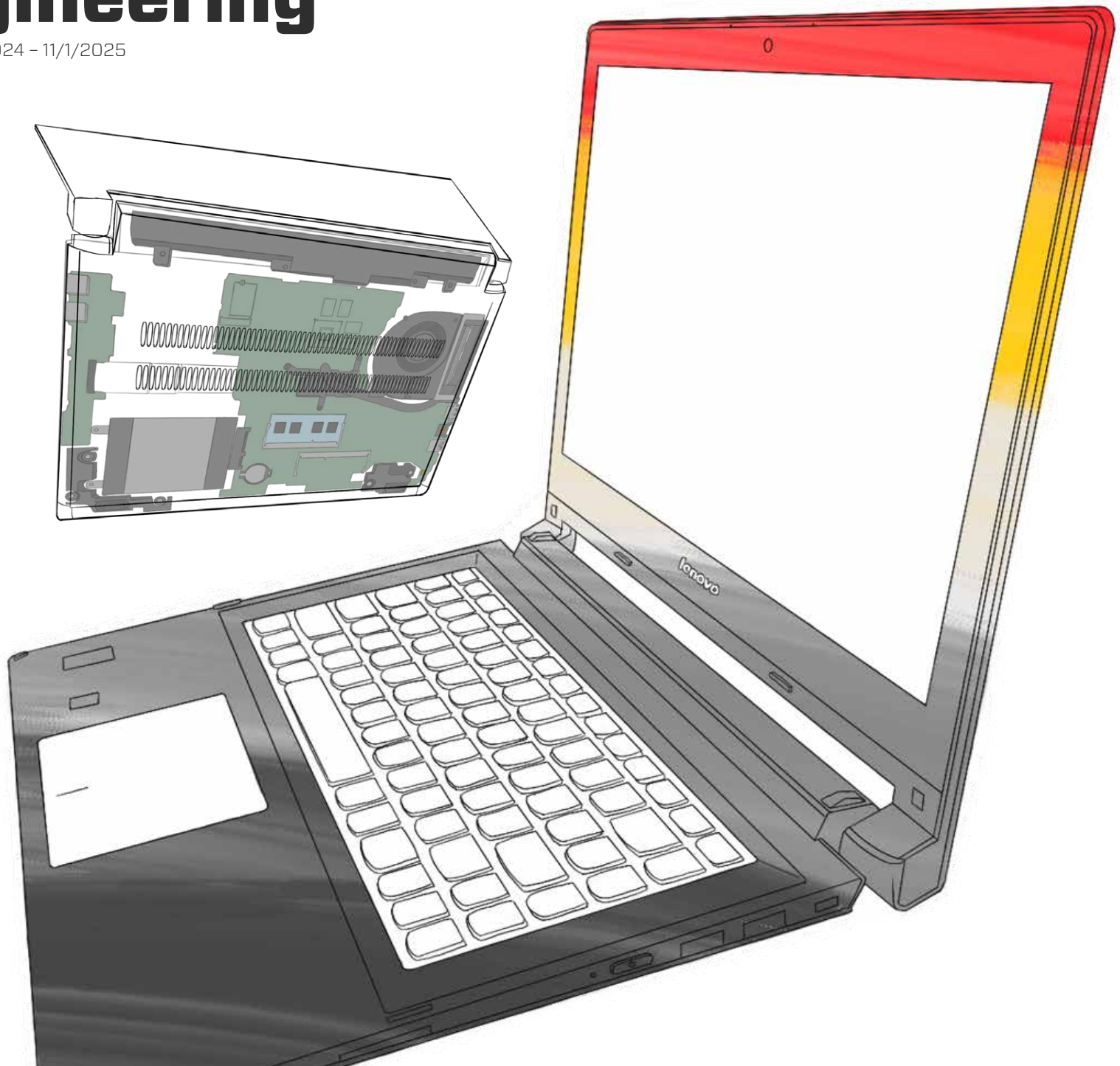
Reverse Engineering

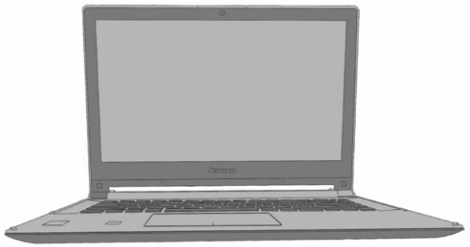
UNIVERSITY OF BATH | **INDIVIDUAL PROJECT** | 4/10/2024 – 11/1/2025

This project entailed a thorough review of an existing product to gain insights as to why it was designed the way it was, including deep dives into the inner workings and suggestions of potential improvements.

The review covers user experience, component functions, manufacturing, materials and cost. A Lenovo Flex 2-14 laptop was reviewed in this work.







Overview

Mid-Range, And More

The Lenovo Flex 2-14 is part of Lenovo's 2014 mainstream lineup of laptops. As such, it doesn't boast the powerful processing power that flagship laptops do, but it makes up for what it lacks with a lower price tag and additional features to boot. **Salient features** include it's 2-in-1 foldable panel (up to a maximum of 300°) and a touchscreen display.

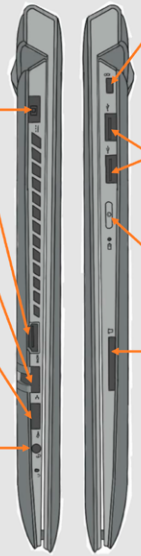
Robust But Hefty

Out of the box, this laptop feels like it can withstand being jostled around in a backpack. Instead of using lightweight metal alloys, the Flex 2-14 uses ABS plastic for its chassis. Clocking in at about 1.9 kg, its on the heavier side, but this choice makes it cheaper, and more damage proof than a large percentage of laptops on the market.

Usability

The ports for the Lenovo Flex 2-14 were carefully selected to widen its market appeal

Port Selection



Charging Port
Standard yellow square-tip charging port

HDMI Port
A useful port for presenting in an **office** meeting or streaming media.

Ethernet Port
Ethernet allows a direct connection to an **office** or **school** network.

USB 3.0 Port
Useful for transferring large amounts of data from an external SSD or HDD quickly.

Headphone Jack
Positioning seems **counter-productive**, given the port is further away from the user in Stand Mode, designed for consuming media.

Kensington Lock Slot
This slot acts as a **security measure**, allowing users to physically lock the laptop to a desk. This proves useful in an **office** setting, preventing loss of sensitive data.

2x USB 2.0 Port
Useful for plugging in mice, **game controllers**, **printers**, **hard keys** for two factor authentication, and more.

Power On Switch
The power switch has an **obscure position**, but the user quickly becomes accustomed to it.

SD Card Reader
A handy tool for a variety of purposes, enabling:
- **Photographers** to upload photos and videos to their system
- **Gamers** to back up save files from their consoles

Usability

The tightness of the hinge, while useful for keeping the screen upright, has its drawbacks

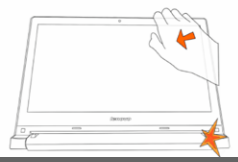
General Use

Not So Seamless
The laptop cannot be opened with one hand. The hinges are so tight that lifting the top panel lifts the entire machine off the table. The user must use their free hand to push down on the keyboard deck to open it.



Ergonomic Trackpad Placement For the 85th Percentile

The trackpad is shifted towards the left, aligning with the right-handed user's natural arm position, reducing strain and making it easier to access while typing. This in turn makes it harder for the left-handed user to access the trackpad.



Less 'Flex' Than Before

Among users familiar with Lenovo's Yoga 2 13 from the year prior to the release of the Lenovo Flex 2-14, some reported broken hinges due to them pushing the panel beyond it's 300° limit. This could be caused by a misconception that the Flex 2-14 would have a 'tablet mode' similar to that of the Yoga 2 13.

Styling

A Pop Of Colour

This laptop was made available in a variety of colours including Red, Yellow, White, Grey and Black.

The message here is likely along the lines of "sleek or stylish, we have you covered" and it goes beyond just the look. This choice in colour schemes portrays the laptop as a versatile machine for both work and play.



Deceptive Borders

The edges of the Flex 2-14 are filleted to depict a slimmer frame when viewed from the front and sides. It is only when the laptop is viewed from the back does the real thickness become apparent.

In most cases, users correlate slimness with a higher quality laptop.

This is likely due to Apple's minimalist and highly premium line of MacBook laptops.



Broad Target Market

The Lenovo Flex 2-14 is a budget-friendly laptop mainly targeting students and office workers, but it appeals to more than just those two demographics.

Bridging the Gap

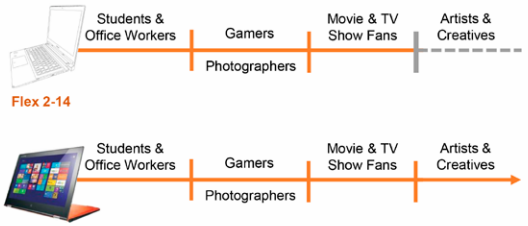
- A good selection of IO ports
- Combined with 2-in-1 and touchscreen functionality
- And a sleek yet sturdy chassis with a variety of colours to choose from



Leads to a versatile laptop that suits needs and preferences of a large number of users

Lenovo Flex 2-14 (2014) vs Yoga 2 13 (2013) Market Appeal

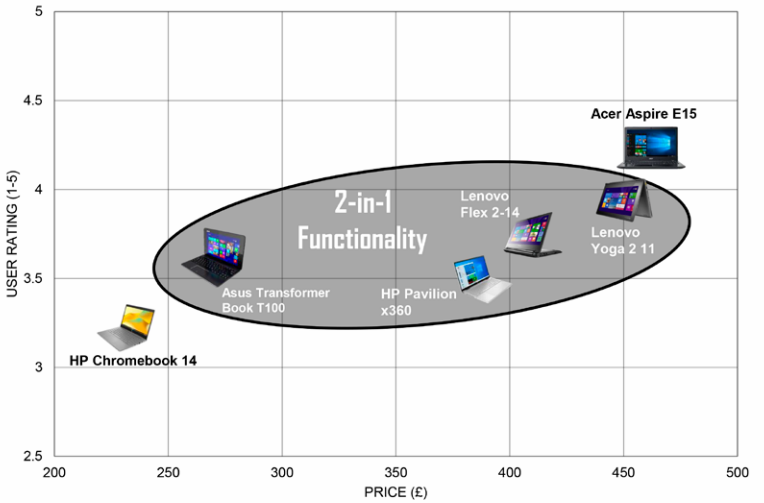
[Even though the Yoga 2 13 was released a year before the Flex 2-14, Its important to remember that it is part of Lenovo's High-End lineup therefore costs more, whereas the Flex 2-14 is a Mid-Range laptop.]



Market Competitors

User Rating vs Price Graph for Budget Laptops

Entry-Level to Mid-Range



Data gathered from Laptopmag.com and amazon.co.uk
Images of products obtained from google.com

Flex 2-14

Analysis

+50% price for +3% Satisfaction?

When comparing the Lenovo Flex 2-14 to the Asus Transformer Book T100, It becomes clear that the difference in user ratings doesn't justify the difference in price point.
This can be partially attributed to the newer and faster processor in the Flex 2-14, but that alone can't explain the massive disjoint between price and user satisfaction.

Only Halfway There

Every laptop in the 2-in-1 Functionality bubble, besides the Flex 2-14, is capable of a full 360° rotation around its hinge.



Lenovo Yoga 2-11

As such, the '2-in-1' functionality of the Flex 2-14 leaves something to be desired

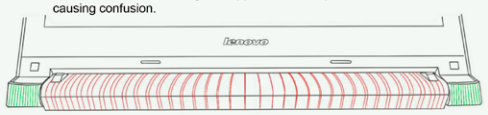
Improvements

The key inspiration for these improvements comes from users who unknowingly pushed the display panel on the Flex 2-14 beyond its 300° limit

Upgrades To Hinge

Solution 1 – Visual Feedback

When the hinge has reached its rotational limit, the base of the hinge protrudes further than the joint pieces. Since the joint pieces aren't completely flat and have a filleted edge, it may lead some users to believe that the filleted edge is supposed to line up with the base, causing confusion.

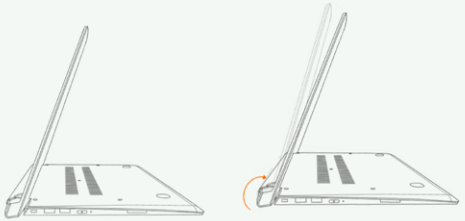


To combat this problem, the base plate can be chamfered such that it aligns perfectly with the joints once it's reached its maximum rotational limit.
This provides a form of visual feedback to the user, informing them that they have pushed the display panel as far back as it can go.



Solution 2 – Snapping

Another solution to the problem is to incorporate an automatic snapping mechanism when the panel reaches its rotational limit. As the hinge approaches 300°, it automatically snaps in place, in much the same way it does when it approaches 0° (closed state).
This doubles as auditory feedback, making certain that the user understands that it has reached its limit.



Component Role And Function

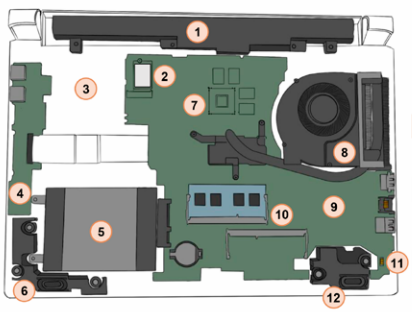
lenovo

Flex 2-14 20404
2-in-1 Laptop

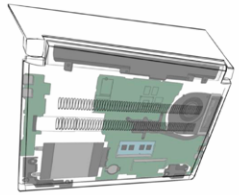
Good Use of Space

Inside the Flex 2-14 is a selection of parts that are neatly arranged and well secured.
The image below is of the underside of the laptop.

Internals



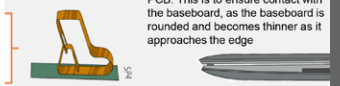
1. **7.5V Battery** — placed inside the hinge-base.
2. **Wi-Fi Card**
3. **Empty Space** — Strategic use of empty space to encourage air flow through the chassis. Also acts as a thermal buffer, preventing heat transfer to areas that the Heatsink can't influence.
4. **Minor PCB** — To connect USB ports and power on button to the main board.
5. **Hard Disk Drive**
6. **Speaker (R)**
7. **Discrete GPU slot** — For higher end models.
8. **Cooling Fan** — Prevents heat buildup in CPU die. Expels internal heat to environment.
9. **Major PCB** — Main board for most component connections.
10. **2x RAM Slot** — Dual-Channel RAM is proven to be the fastest configuration.
11. **EMI Shield** — Prevents Electro-Magnetic Interference throughout the PCB
12. **Speaker (L)**



Closer Look

EMI Shield

The EMI Shield is strategically placed in the furthest corner of the PCB. This is to ensure contact with the baseboard, as the baseboard is rounded and becomes thinner as it approaches the edge



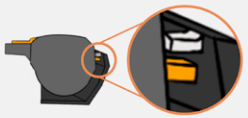
This component prevents Electro Magnetic Interference by grounding the PCB to a slab of metallic tape on the baseboard. When contact is established, the metallic strip breaks its short circuit and connects the PCB to ground. This acts somewhat like an inverted switch.

7.5V Battery

The Battery supplies power directly to the PCB.

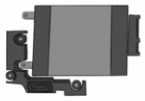


Since the battery is flipped upside down during regular use, there are added fins on the pack that rest on fins coming out of the hinge
base to prevent a large bending moment across the cross section of the battery.

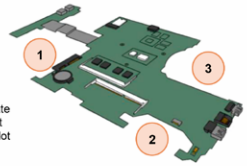


PCB Built Like A Jigsaw Puzzle

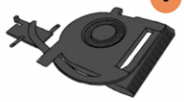
1



PCB is cut out to fit perfectly in the chassis while accommodating all the components that connect to it.

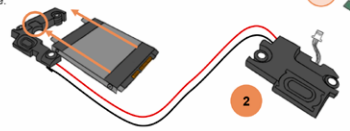


3



HDD (Hard Disk Drive) connects to the PCB by sliding into its slot. To accommodate for this, the adjacent speaker has an indent to allow the HDD to slide in and out of its slot with ease.

Cooling Fan is placed as close to the processing units as possible to increase cooling efficiency. The PCB is cut out to allow placement.



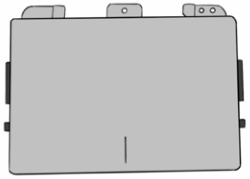
Speakers Both connect to the same pin on the PCB. It is cut to allow them to be placed on both sides of the laptop, increasing sound quality

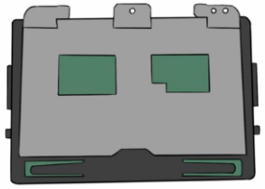
Peripherals

Accurate Tracking

The Trackpad on the Flex 2-14 incorporates a capacitive sensing layer to precisely locate the placement of one's finger.
This works by calculating the change in capacitance of electrodes placed on the plane below the touch layer.

This trackpad is also capable of sensing gestures such as a two-finger stroke being bound to page up and page down.





Tactile But Tedious

This trackpad uses two cantilever beams to detect clicks without making two separate touch layers to interact with. This is done by triggering mechanical switches after enough force is applied to a particular region of the trackpad.

A drawback to this is in that the user must press down with enough force to trigger these switches every time they need to click, which gets tedious. The trackpad is capable of recognising a click from a tap, but the software tends to get it confused with a resting finger on the pad.

Funnel For Liquids... Helps?

The keyboard deck of the flex 2-14 has an indent for the placement of the keyboard. This indent acts as a funnel for liquids to seep into the keyboard.

Surprisingly, this is actually a safety feature.



The bottom of the keyboard is sealed with a metallic sheet, protecting internal components from water damage in the event of an accidental spill.

Since the keyboard pops right out of the laptop, it can be easily and cheaply replaced once damaged, unlike the expensive internals.



lenovo

Flex 2-14 20404
2-in-1 Laptop

Component Role And Function

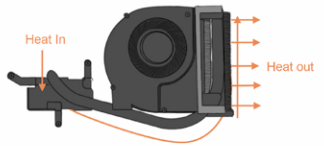
Deep Dive

The Cooling System of the Lenovo Flex 2-14 comprises of a cooling fan, heatsink and heatpipe.

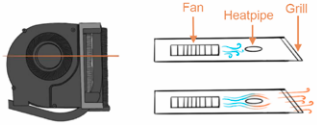
Cooling

How Does It Work?

The heat from the CPU travels to the Heatsink. Contact between these two surfaces is maximized by adding **Thermal Paste** removing air gaps that act as insulators.
The heat is then pulled from the heatsink into the heatpipe, which is cooled by the cooling fan.
Finally, heat is expelled from the system via grill fins.



Cross Section View



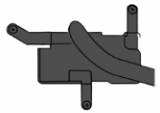
The heatpipe is shaped similarly to an airfoil. It allows for an even flow of cool air across it to cool it down.

The warm air from across the heatpipe is pushed further down into the grill. The grill has several fins that direct the warm air outwards to the surroundings.

Heat Sink-Spreader Contact

It is vital that contact between the CPU heatspreader and the heatsink is maximized. This is more so the case, since the PCB is upside down during normal use, so more care needs to be put into the mounting of the heatsink.

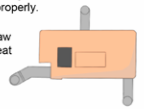
The screws to tighten the mount of the heatsink are strategically placed in a triangle around the CPU. This configuration evenly distributes the pressure across the entire CPU heatspreader.



No Contact With GPU Die

The integrated GPU in the Flex 2-14 is extremely low in performance. As such, it produces very little heat and does not require a heatsink to function properly.

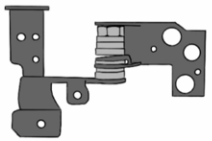
The heatsink is indented to withdraw from the GPU die, such that the heat from the CPU doesn't travel to it and reduce performance further.



Deep Dive

The Hinge mechanism in the Lenovo Flex 2-14 is surprisingly standard, given its flex functionality. Standard or not, the hinge has an inspired method of closing the laptop lid.

Hinge



The Mechanism

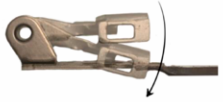
The hinge mechanism is built using good quality metal parts. This is important as it will be used constantly over a period of 4-7 years. The hinge barrels fit tightly onto the hollow inner tube, creating high friction. This is required to secure the laptop screen in any position the user may prefer, and is especially important in the case of the Flex 2-14 with it's 2 in 1 functionality.

Lacking In Novelty

It's disappointing to open up a laptop branded as a 2-in-1 to find a standard laptop hinge.
This is likely the reason that the Flex 2-14 can only rotate 300° instead of 360°, because it was simply cheaper and easier for Lenovo to use a standard, widely available laptop hinge rather than manufacturing a new one.

Snap-Shut Uses Friction

Most laptop lids snap shut automatically after reaching close enough to closed.

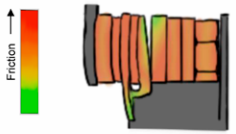


This gives the user feedback that the lid has been successfully closed and the laptop won't open up in their backpack.

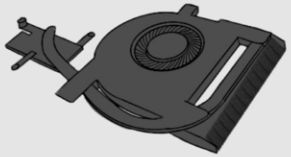
How Does It Work?

Two of the hinge barrels are connected directly to the part of the hinge that mounts on the chassis base. They do not move independently from that part of the hinge.
As such, they create additional friction in the mechanism.

One of these hinge barrels is filleted, such that when the hinge approaches 0°, the additional friction is suddenly removed and the lid snaps shut.







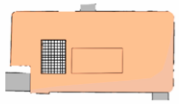
Improvements

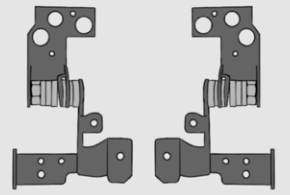
Cooling

Cooler GPU Die

Rather than indenting the heatsink where the GPU die is, an idea to reduce heat and cost is to simply drill straight through that region and add a small dust filter on top.

This allows the hot air trapped in the indent to exit to the surroundings, cooling the GPU die, and potentially having positive impacts on CPU temperatures as well.





Improvements

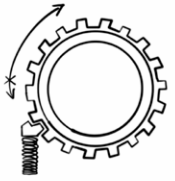
Hinge

Geared Hinge

The hinge of the Flex 2-14 could benefit from having more snap functionality, giving users auditory feedback, letting them know that the lid can no longer be pushed further.

The gear rotates along with the lid. As it rotates, the spring latches onto the next spoke, making a satisfying 'click' sound. The spring stiffness should be calibrated to add enough resistance for stability without making it hard to adjust the lid.

By incrementing the angles of the spokes by 12° each, this hinge allows flexibility in terms of preference alongside functionality.



lenovo

Flex 2-14 20404
2-in-1 Laptop

Manufacturing, Materials & Cost



Overview

The Lenovo Flex 2-14 was designed as a budget laptop, undercutting competitors without compromising in quality. The choices in materials used reflects this goal.

~30% Profit Margin

The retail price of the Lenovo Flex 2-14 is £420.
The estimated cost of all materials and components is approximately £303.
This leaves £117 for manufacturing costs, worker wages and other overheads, with the remainder being profit.

Bill Of Materials

General

Built In-House
Outsourced

No.	Description	Main Material	Cost (£)
1	Top Cover	PC-ABS Plastic	6
2	LCD Screen	Glass + Liquid Crystal	45
3	Hinge x2	Steel	3
4	7.5V Battery	Lithium-ion	40
5	Power Input Cable	Plastic + Copper	2
6	WLAN PCB	FR-4	10
7	Sensor Board PCB	FR-4	8
8	Camera PCB	FR-4	5
9	Major PCB	FR-4	15
10	Minor PCB	FR-4	10
11	CPU: Intel i3 4030U	Silicone	50
12	RAM: DDR3L 8GB	Silicone	25
13	Network Adaptor	FR-4	5
14	Heatsink	Plastic + Copper	10
15	Speaker R	Polycarbonate	4
16	Speaker L	Polycarbonate	4
17	500GB HDD	Aluminium	25
18	Trackpad	FR-4 + Polycarbonate	10
19	Keyboard	FR-4 + Polycarbonate	10
20	Main Chassis	Aluminium	10
21	Bottom Cover	PC-ABS Plastic	6

Bill Of Materials

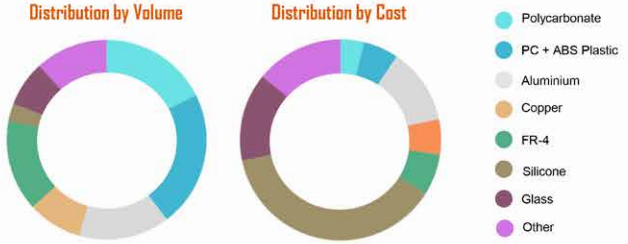
Screws

The laptop has 9 unique screw varieties that are listed below.



No.	Specification	Nylon Coat	Quantity
22	M2 x 6mm silver	Blue	9
23	M2 x 2.5mm silver	N/A	1
24	M2 x 4mm silver	Blue	8
25	M2.5 x 4mm silver	Blue	8
26	M3 x 4mm silver	N/A	4
27	M2 x 3mm black	Blue	3
28	M2 x 3.5mm black	Red	10
29	M2 x 3mm black (6mm Large head)	Blue	2
30	M2 x 3mm silver (8mm Large head)	Blue	6

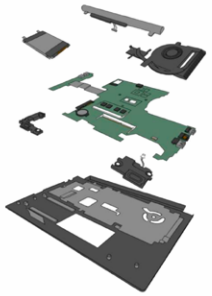
Materials Distribution



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Flex 2-14 20404
2-in-1 Laptop
Manufacturing, Materials & Cost

Deep Dive: Assembly Hiccups

Assembly

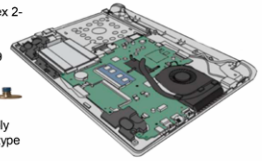


Simplicity Screwed

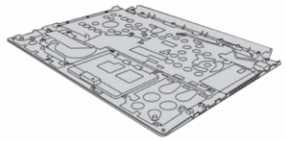
On the surface, the assembly of the Flex 2-14 seems simple and similar to most laptops.
This is heavily impacted by the use of 9 different varieties of fasteners.



25 minutes were wasted in the assembly process due to using the wrong screw type to fasten the Major PCB to the chassis. This was only realised when attempting to fasten the bottom cover using the very similar but 2mm shorter silver screws, which were unable to reach the threads deeper in the chassis.

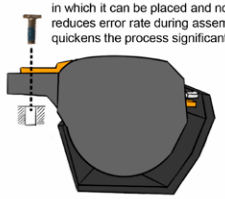


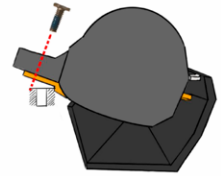
Standardizing fasteners with a maximum of 3 or 4 unique varieties can significantly reduce the error rate of the assembly.



Poka-Yoke

The battery pack has a unique orientation in which it can be placed and no other. This reduces error rate during assembly and quickens the process significantly.





Disassembly Made Difficult

This laptop has an unorthodox method of disassembly. Taking the keyboard off is the first step to opening it, after which screws must be removed from the keyboard deck that then allows the bottom cover to come loose. Removing the keyboard requires a wedging tool, which most customers aren't equipped with.
This is likely done to discourage DIY servicing of the laptop and instead sending it to Lenovo to fix for a price. This could also be a way to void the warranty in case the customer damages the product in an attempt to service it.



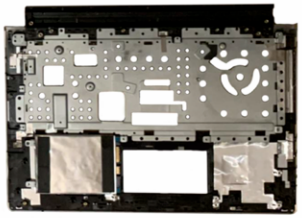
One And Only Assembly

The service manual of the Flex 2-14 describes a Nylon coat used on the screws. It states that
"Each screw should only be used once".
This is to avoid wear of the nylon coats. Once worn, the screws may come loose due to vibrations in the system.
The only screws with no nylon coat are those securing the Hard Disk Drive, which is often removed / replaced.

Deep Dive: Chassis Manufacturing Processes

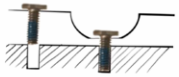
Manufacturing

Several manufacturing processes have been used to produce the shape of this chassis. These are discussed below.



Stamping/Punching – circular and rectangular holes on aluminium plate for reduced weight and access to cables from keyboard deck.

Deep drawing – to reduce distance between screw and hole, reducing required size of screw and variety of screws necessary.



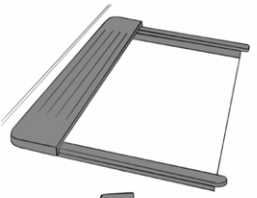
Complex Moulds

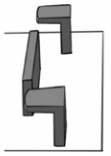
The plastic portion of the chassis used Injection Moulding to be produced. The chassis had a number of overhangs, leading to more complex moulds.

The trackpad housing protrudes outward with a cavity underneath, suggesting a mould with a slider.

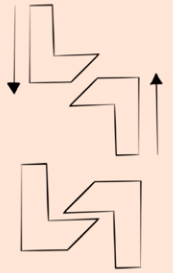
Additionally, there were small lips meant to keep the speaker wiring in place. These also had overhangs that most likely used a sliding mould to accomplish.

Finally, the protrusions for Snap Fitting to the Bottom cover also included lip-like features, complicating the mould further.





Improvement: Chassis-Cover Interface

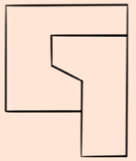


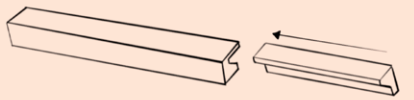
The current Chassis-Cover Interface uses several lip-like features on both surfaces to snap fit to each other. This requires significant complexity in the moulds as these features are too small to be moulded separately and then placed onto the chassis/case while maintaining structural integrity after snap fitting.

Instead, it's likely that the mould involves several sliders that move out of the way when the part cools down to allow it to be lifted out.

Using a slider instead of snap fitting alleviates this issue for a number of reasons:

1. Unlike Snap-Fitting, the lips do not suffer impacts, and can be moulded separately and then placed onto the chassis/cover without worry of failure.
2. When moulded, the slight angle on the part allows it to be pulled out at an incline without the need to slide out a part of the mould, significantly reducing complexity
3. A slider would make disassembling the laptop far easier and less risky to the internal components as inexperienced users may wedge a metal tool to prop open the snap-fitted cover and potentially damage parts.





Drawing

SKETCHING, DRAWING AND ILLUSTRATION

